EXPERIMENTAL PAPER BLUEPRINT

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Programme & Institution:	MSc Data Science and Computational Intelligence | Softwarica College / Coventry University
Module & Assignment:	STW7072CEM / ST7072CEM Machine Learning Coursework CW (Regular)
Coursework Research Project:	**Machine Learning-Based Analysis of Daily Rainfall Patterns and Extreme Rainfall Events in the Kathmandu Valley**

Reference Paper Blueprint

- **Target Paper:** 'Evaluation of IMERG Precipitation Product Downscaling Using Nine Machine Learning Algorithms in the Qinghai Lake Basin'
- **Citation:** Lei et al. (2025), *Water*, 17(12), 1776.
- Paper link: **https://doi.org/10.3390/w17121776**

How to use this worksheet

1. Check the average length of comparable papers on a similar topic or in the same journal—look for papers close to yours in methodology.

Comparable paper	Length (excl. references)
Paper 1: Lei et al. (2025), Water 17(12), 1776	~8,400 words
Paper 2: Sharma et al. (2020), Earth and Space Science 7(8), e2020EA001315	~7,900 words
Paper 3: Khanal et al. (2024), J. Hydrol. Meteorol. 12(1), 34-45	~4,200 words
Paper 4	—
Paper 5	—
Average	**~6,800 words**

2. Adjust the overall suggested length using that average.

3. Adjust the length of each section based on the papers you reviewed.
4. Check which sections are used in your field and edit the table accordingly:
    - Which sections from the table are present? Delete any that aren't.
    - Is there a separate literature review section?
    - Are results and discussion one section, or two?
    - Are discussion and conclusion one section, or two?
5. Compare the elements listed in each section against papers in your field:
    - Are all the listed elements present? Delete any that aren't.
    - Are they presented in the same order? If not, reorder the table.
6. Save the completed table below as its own document—this becomes your personal blueprint for future empirical papers.

Blueprint

Target Length Specifications

- **Benchmark:** 6,000–10,000 words (without references)
- **Recommended range:** 7,500–9,500 words
- **Average length in target field:** ~8,400 words

Section	Element	Purpose	In my field?	My notes and ideas
Introduction*(800–1200 words)*	Importance of the topic	Introduce the topic and outline why it should be studied	✓ Yes	High-resolution **precipitation** datasets are vital for alpine hydrological modeling, regional water resources management, and **climate** attribution. Coarse satellite grids (0.1°) smooth out critical **precipitation** gradients in complex mountain basins.
	Brief literature review	Situate your study within the context of your field	✓ Yes	Reviews dynamical vs. statistical **downscaling**; notes shift from traditional regression (MLR, GWR) to **machine learning** (RF, **SVM**, **ANN**, **XGBoost**) and **deep learning** architectures (**CLSTM**, **Transformer**, **GAN**); highlights limited comparative evaluations in high-altitude plateau catchments.
	Research gap	Provide justification for your study	✓ Yes	(1) Lack of systematic intercomparison between traditional **ML** and modern deep-learning architectures

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				(**Transformer**, **GAN**) in alpine basins; (2) Unclear necessity, **spatial** thresholding, and risk of residual correction (e.g., noise amplification / mode collapse).
	Research aim	Present what your study was about	✓ Yes	Systematically evaluate nine **ML**/DL algorithms (**ANN**, **CLSTM**, **GAN**, **KNN**, MSRLapN, RF, **SVM**, **Transformer**, **XGBoost**) to downscale **IMERG** V07 from 0.1° to 0.01° in the Qinghai Lake Basin (2001–2020), validating against ground rain gauges and examining residual correction.
	The main contributions of the paper *(optional)*	Highlight what novel insights your study offers	✓ Yes	Generates a 20-year, 0.01° gridded **precipitation** dataset; identifies **KNN** as optimal for **spatial** gradient capture; evaluates seasonal performance (cold vs. warm); provides **SHAP**-based interpretability of environmental drivers (NDVI, lat, lon).
	Structure of the paper *(optional)*	Present the organization of the paper	✓ Yes	Standard scientific IMRAD: Sec 2 describes study basin, multi-source datasets, 9 **ML** algorithms, and residual correction workflow; Sec 3 details **validation** results across timescales; Sec 4 discusses **downscaling** value, residuals, and **SHAP**; Sec 5 concludes.
Literature Review*(optional 1000–1500 words)*	Key themes from the literature / Definitions of key concepts	Outline the literature on the main topics related to your research	✓ Yes	In MDPI Water and hydrology journals, literature review is integrated directly into Introduction (Sec 1) and Discussion (Sec 4) rather than presented as a standalone section.
	Critical analysis	Analyze previous studies critically to reinforce the research gap	✓ Yes	Critiques conventional **ML** studies that overlook **spatial** detail attenuation in deep nets or uncritically apply residual correction without checking error threshold bounds.

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Theoretical / Methodological Framework*(optional 200–500 words)*	The theories utilized in the study	Present which theories you drew upon when designing the study and/or analyzing the data	✓ Yes	Statistical **downscaling** principles (modeling empirical relationships between coarse **rainfall** and high-res physiographic predictors); residual correction theory via bilinear interpolation; game-theoretic **SHAP** feature attribution; K-fold cross-**validation**.
Methodology*(1500–2500 words)*	Definition and justification of the research methodology	Outline and justify the chosen methodology	✓ Yes	Evaluates 9 **ML** algorithms covering diverse paradigms (tree learners, kernel methods, instance-based, shallow/deep neural networks). Combines high-resolution land surface predictors (NDVI, DEM, slope, aspect, coordinates) with residual error correction.
	Background information about the context of your study *(optional)*	Present the context in which your study was conducted (e.g., geography)	✓ Yes	Qinghai Lake Basin (36°32'–37°15'N, 99°36'–100°47'E), NE Qinghai-Tibet Plateau. Elevation 3176–5178 m; alpine semi-arid plateau continental **climate**; distinct cold (solid) and warm (liquid) seasons; expanded rectangular domain to ensure sufficient 0.1° training pixels.
	Sample and sampling techniques + ethical considerations	Outline who or what you studied, how you got it, and any ethical considerations	✓ Yes	26 rain gauge stations from 3 networks (20 NMIC daily gauges, 9 high-frequency Heihe gauges, 2 wetland stations). **GPM IMERG** V07 daily 0.1° data (2001–2020). MOD13A3 NDVI (250m) and SRTM DEM (90m). Purely physical/meteorological data.
	Research tools and procedures	Specify which research tools you used and specifically how you conducted your study	✓ Yes	Python, ArcGIS 10.7, PyTorch/Scikit-Learn/**XGBoost**. Procedures: Resampling predictors to 0.1° and 0.01°; training 9 **ML** models at 0.1°; predicting 0.01° **precipitation**; calculating 0.1° residuals; bilinear interpolating residuals to 0.01°; adding residuals to outputs.
	Data analysis techniques	Present how you analyzed your data and justify your choice	✓ Yes	Model reconstruction checked via 5-fold CV (R^2, Bias, MAE, CRMSE, RMSE, Sigma Ratio). Downscaled gridded

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				evaluation against 26 gauges via CC, RMSE, MAE, R^2, Sigma Ratio, KGE, and Bias. Seasonal evaluation via Taylor diagrams and heatmaps. **SHAP** feature importance.
Results*(2000–3500 words)*	Presentation of the main results	Present the results you obtained	✓ Yes	(1) 0.1° reconstruction: **XGBoost** and RF lead (R^2=0.97 and 0.94); (2) 0.01° downscaled gauge **validation**: **KNN** performs best post-calibration (KGE=0.70, RMSE=17.09mm, Bias=-3.31mm), followed by **Transformer** (KGE=0.69); (3) **GAN** degrades post-calibration (KGE=0.61); (4) Cold season: **KNN**/**ANN** lead; Warm season: **Transformer**/**KNN** lead.
	Clear use of tables, figures or quotes	Provide a visual representation of the data and raw evidence	✓ Yes	Fig 1: Study area & station map; Fig 2: **Downscaling** workflow; Fig 3 & 4: 0.1° metrics & **spatial** maps; Fig 5 & 6: 0.01° **precipitation** maps pre/post calibration; Fig 7: Metric boxplots; Fig 8 & 9: Taylor diagrams and seasonal heatmaps; Fig 10 & 11: Quantile error plots; Fig 13: 20-yr mean residual maps.
Discussion*(1200–2000 words)*	Restatement of the result to be discussed	Remind the reader of the result	✓ Yes	Highlights **KNN**'s overall superiority in capturing orographic **precipitation** gradients and low-intensity events; notes **Transformer**'s competitive stability but smooth **spatial** attenuation; summarizes **GAN**'s failure.
	Comparison of the result with the literature	Situate the result within the literature context	✓ Yes	Compares tree-ensemble stability with prior regional **downscaling** studies; discusses **deep learning**'s parameter sensitivity; contrasts findings with studies advocating unconditional residual correction.
	Explanation of the findings and/or differences with literature	Explain the findings you obtained and possible differences with previous studies	✓ Yes	**KNN** succeeds due to non-parametric local feature weighting; **Transformer** smooths local peaks due to global self-attention; **GAN** fails because bilinear

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				interpolation amplifies generator noise across complex terrain (mode collapse). Residuals concentrate along the lakeshore due to sand/anthropogenic distortion of NDVI.
	Interpretation of the findings	Show what your findings mean and how they can be understood	✓ Yes	High-resolution **precipitation** fields resolve microclimate gradients essential for alpine water resource management. Residual correction is model-specific and requires noise threshold safeguards.
Conclusion*(600–1000 words)*	Restatement of the main aim and topic	Remind the reader of the aim and topic of your study	✓ Yes	Synthesizes the systematic **downscaling** of **GPM IMERG** V07 across the Qinghai Lake Basin using nine **ML**/DL paradigms.
	Recap of the key findings and contributions + brief discussion	Restate the main results and discuss briefly	✓ Yes	Calibrated **KNN** delivers the most faithful **spatial** gradients (KGE=0.70); residual correction benefits 8 models but fails for **GAN**; NDVI, longitude, and latitude are dominant predictors according to **SHAP**.
	Practical implications of the findings	Show what impact your findings can have for different stakeholders	✓ Yes	Supplies a validated 0.01° **precipitation** dataset for Tibetan Plateau hydrological models, ecological conservation, and **climate** change monitoring.
	Limitations	Acknowledge the limitations of your study	✓ Yes	Gauge scarcity in high-altitude zones; monthly **temporal** scale overlooks sub-daily flash storms; poorer satellite performance over open water bodies.
	Suggestions for future research	Suggest what future research should focus on	✓ Yes	Incorporate physically constrained interpolation for GANs; test higher **temporal** resolutions (hourly/daily); integrate additional atmospheric variables and test multi-model ensemble stacking.